

Research Notes on Generative adversarial network

Generative adversarial network

$$\min_G \max_D E[\log D(x)] + E[\log(1 - D(G(z)))]$$

>> Section 1

Generative reparametrization The GAN architecture has two main components. One is casting optimization into a game, of form $\min_G \max_D L(G, D)$, which is different from the usual kind of optimization, of form $\min_{\theta} L(\theta)$. The other is the decomposition of μ_G into $\mu_Z \circ G^{-1}$, which can be understood as a reparametrization trick. To see its significance, one must compare GAN with previous methods for learning generative models, which were plagued with "intractable probabilistic computations that arise in maximum likelihood estimation and related strategies". At the same time, Kingma and Welling and Rezende et al. developed the same idea of reparametrization into a general stochastic backpropagation method. Among its first applications was the variational autoencoder.

>> Section 2

Generator-encoder team aims to minimize the objective, and discriminator aims to maximize it: $\min_{G, E} \max_D L(G, E, D)$. In the paper, they gave a more abstract definition of the objective as: $L(G, E, D) = E(x, z) \sim \mu_E, X[\ln D(x, z)] + E(x, z) \sim \mu_G, Z[\ln(1 - D(x, z))]$ where $\mu_E, X(dx, dz) = \mu_X(dx) \cdot \delta E(x)(dz)$ is the probability distribution on $\Omega_X \times \Omega_Z$ obtained by pushing μ_X forward via $x \mapsto (x, E(x))$, and $\mu_G, Z(dx, dz) = \delta G(z)(dx) \cdot \mu_Z(dz)$ is the probability distribution on $\Omega_X \times \Omega_Z$ obtained by pushing μ_Z forward via $z \mapsto (G(x), z)$. Applications of bidirectional models include semi-supervised learning, interpretable machine learning, and neural machine translation.

>> Section 3

Compared to fully visible belief networks such as WaveNet and PixelRNN and autoregressive models in general,

GANs can generate one complete sample in one pass, rather than multiple passes through the network. Compared to Boltzmann machines and linear ICA, there is no restriction on the type of function used by the network. Since neural networks are universal approximators, GANs are asymptotically consistent. Variational autoencoders might be universal approximators, but it is not proven as of 2017.

>> Section 4

Other uses Other than for generative and discriminative modelling of data, GANs have been used for other things. GANs have been used for transfer learning to enforce the alignment of the latent feature space, such as in deep reinforcement learning. This works by feeding the embeddings of the source and target task to the discriminator which tries to guess the context. The resulting loss is then (inversely) backpropagated through the encoder.

>> Section 5

Equilibrium when discriminator moves first, and generator moves second: $\mu^D \in \arg \max_{\mu^D} \min_{\mu^G} L(\mu^G, \mu^D)$, $\mu^G \in \arg \min_{\mu^G} L(\mu^G, \mu^D)$, $\hat{\mu}_D \in \arg \max_{\mu^D} \min_{\mu^G} L(\mu^G, \mu^D)$, $\hat{\mu}_G \in \arg \min_{\mu^G} L(\mu^G, \mu^D)$.

>> Section 6

Equilibrium when generator moves first, and discriminator moves second: $\mu^G \in \arg \min_{\mu^G} \max_{\mu^D} L(\mu^G, \mu^D)$, $\mu^D \in \arg \max_{\mu^D} L(\mu^G, \mu^D)$, $\hat{\mu}_G \in \arg \min_{\mu^G} \max_{\mu^D} L(\mu^G, \mu^D)$, $\hat{\mu}_D \in \arg \max_{\mu^D} L(\hat{\mu}_G, \mu^D)$.

>> Section 7

The discriminator's strategy set is the set of Markov kernels $\mu_D : (\Omega, \mathcal{B}) \rightarrow P([0, 1], \mathcal{B}([0, 1]))$, where $\mathcal{B}([0, 1])$ is the Borel σ -algebra on $[0, 1]$. Since issues of measurability never arise in practice, these will not concern us further.

>> Section 8

(Ω_Z, μ_Z) , the latent space. There are 3 players in 2 teams: generator, encoder, and discriminator. The generator and encoder are on one team, and the discriminator on the other team. The generator's strategies are functions $G : \Omega_Z \rightarrow \Omega_X$, and the

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encoder's strategies are functions $E : \Omega_X \rightarrow \Omega_Z$ $\{\displaystyle E:\Omega_X\to \Omega_Z\}$. The discriminator's strategies are functions $D : \Omega_X \rightarrow [0, 1]$ $\{\displaystyle D:\Omega_X\to [0,1]\}$. The objective function is $L(G, E, D) = \mathbb{E}_{x \sim \mu_X} [\ln D(x, E(x))] + \mathbb{E}_{z \sim \mu_Z} [\ln (1 - D(G(z), z))]$ $\{\displaystyle L(G,E,D)=\mathbb{E}_{x\sim \mu_X}[\ln D(x,E(x))]+\mathbb{E}_{z\sim \mu_Z}[\ln(1-D(G(z),z))]\}$

>> Section 9

Medical One of the major concerns in medical imaging is preserving patient privacy. Due to these reasons, researchers often face difficulties in obtaining medical images for their research purposes. GAN has been used for generating synthetic medical images, such as MRI and PET images to address this challenge. GAN can be used to detect glaucomatous images helping the early diagnosis which is essential to avoid partial or total loss of vision. GANs have been used to create forensic facial reconstructions of deceased historical figures.